Flame Tests: Identifying Metal Ions by Color

U2 / S2.5

BEFORE THIS IS COPIED — OPEN ITEMS

- **Draft.** A flame-test lab at S2.5 is not yet a confirmed lab placement in courses/chemistry/DECISIONS.md. It becomes final once you confirm it.
- **Iron nitrate.** Your list reads "$FeNO_3$ (Fe^{+2})". The iron(II) compound is $Fe(NO_3)_2$, which is what the handout prints. Most stockrooms carry iron(III) nitrate, $Fe(NO_3)_3$, instead. Check the bottle and change the name and formula at station 5 to match it.
- **Cobalt(II) chloride.** Its SDS hazard statements need checking before students handle it; cobalt compounds are commonly classed as sensitizers and carcinogens. If the SDS confirms that, swap it for a compound already in the set, or run cobalt as a teacher demonstration.
- **Heat source.** The handout assumes a Bunsen burner at each of the six stations.
- **Every hazard and disposal statement** on the handout is marked not yet verified below. Check each against the SDS and district policy and date it.

PREP — THE DAY BEFORE

WHEN	TASK	DETAIL
Day before	Check stock solutions	Use the solutions from previous years. Record each concentration here so the next batch matches: _____. Top up any cup below about 1 cm deep.
Day before	Make the unknowns	Pour a sample cup of each and label only the letter: **A = copper(II) nitrate · B = strontium chloride · C = potassium nitrate** (your choice of copper, strontium and potassium).
Morning of	Load stations	Six stations, each with a burner, a striker, two labeled cups (layout below) and a cup of clean sticks. Sticks per period = students × 12 tests.
Before 1st period	Room	Clear the sinks so used sticks can drop in. Notebooks stay closed and away from flames except when recording.

Station layout

STATION	SAMPLE 1	SAMPLE 2
1	Lithium chloride, $LiCl$	Sodium nitrate, $NaNO_3$
2	Potassium nitrate, KNO_3	Calcium nitrate, $Ca(NO_3)_2$
3	Strontium chloride, $SrCl_2$	Barium nitrate, $Ba(NO_3)_2$
4	Copper(II) nitrate, $Cu(NO_3)_2$	Cobalt(II) chloride, $CoCl_2$
5	Iron nitrate (confirm which, above)	Unknown A — copper(II) nitrate
6	Unknown B — strontium chloride	Unknown C — potassium nitrate

TIMING — 50-MINUTE PERIOD

MIN	PHASE	WATCH FOR
0–5	Pre-lab check, goggles on, hair tied	No pre-lab, no burner.
5–8	Part A: light burners at the first station	Yellow, smoky flames: open the air.
8–38	Part B and C: six rotations of 5 minutes	Call "rotate" on the minute. Groups that finish early wait; they do not drift.
38–45	Finish the table, compare with the group	Colors written from memory.

45–50	Cleanup, goggles off last	Sticks in the trash instead of the sink.

FAILURE POINTS

- **Everything looks yellow.** Sodium is everywhere, and the burning wood is yellow-orange. Say: "Record the first two seconds, then the stick goes in the sink."
- **Potassium's lilac is faint.** Any trace of sodium swamps it. Students who see yellow at station 2 or for unknown C usually re-dipped a stick or touched the tip.
- **Re-dipped sticks.** One dirty stick contaminates a cup for every later group. Watch the first rotation closely.
- **The stick catches fire.** Usually from holding it in the flame too long. Tip out, into the sink, new stick.
- **"Red" for three compounds.** Push for crimson, bright red or orange-red. Lithium, strontium and calcium are where students lose the unknown.

EXPECTED RESULTS

A good group records the reference color for all seven ions on the table and names all three unknowns. A group that records "red" without more detail still gets A and C, but cannot separate B from lithium chloride: that is exactly post-lab question 2.

SAMPLE	EXPECTED (S2.5 NOTES)	WHAT STUDENTS OFTEN WRITE
$LiCl \cdot SrCl_2 \cdot Ca(NO_3)_2$	Crimson red · red · orange-red	"Red" for all three
$NaNO_3$	Yellow	Yellow, strong and long-lasting
KNO_3	Lilac	Faint purple, or yellow if contaminated
$Ba(NO_3)_2 \cdot Cu(NO_3)_2$	Pale green · blue-green	"Green" for both; copper is the brighter, bluer one
$CoCl_2$ · iron nitrate	Not on the reference table	Record what your first class sees and add it here: _____

SAFETY + DISPOSAL — NOT YET VERIFIED

- Nitrates as oxidizers (Na, K, Ca, Ba, Cu, Fe nitrate) — check each SDS. Verified on: _____
- Barium, copper and cobalt compounds harmful by swallowing and on skin or eyes — check each SDS. Verified on: _____
- Strontium chloride and calcium nitrate eye hazard — check each SDS. Verified on: _____
- Eyewash duration (15 minutes) — matches the room's standing procedure? Verified on: _____
- Disposal: students pour nothing out; the cups are collected and stored. The end-of-year disposal route for the copper, barium, cobalt and strontium stock solutions follows the district chemical-waste policy — confirm it. Verified on: _____
- Used sticks: collected from the sink after class, drained, and put in the trash.
- Alconox cleanup block is the standing default and needs no per-lab check.

PRE-LAB KEY

1. Ground and excited state. The electron absorbs energy from the flame and jumps from its ground state to a higher level (excited state). It drops back and gives off the energy as a photon of light. Both terms must appear.
2. Same color, $NaNO_3$ and NaCl. The metal ion (sodium) makes the color, not the nitrate or chloride part.
3. Hardest pair. Lithium (crimson red) and strontium (red); calcium (orange-red) is also accepted with one of them. An unknown red could be either, so color alone cannot decide.
4. Part B, step 2. About 1 cm of the tip, for a few seconds; held at the edge of the flame just above the blue inner cone.
5. Part B, step 3. After two or three seconds the wood burns yellow-orange, which is the stick's color, not the solution's.
6. Part B, steps 4–5. It goes in the sink. A used stick carries the first metal into the next cup and contaminates it for every later group.
7. Stick on fire. Take it out of the flame and drop it in the sink; never on the bench or the trash. Full credit needs both room locations written correctly.
8. Data table. Twelve rows in station order, three columns, in ink; expected colors filled from the reference table, with cobalt, iron and the unknowns left blank.

POST-LAB KEY

1. Identify A, B, C. A: copper(II) nitrate, blue-green. B: strontium chloride, red. C: potassium nitrate, lilac. Each needs the color from the student's own table. If a student names lithium chloride for B and backs it with a crimson-red observation, give partial credit and expect question 2 to catch it.
2. Unknown with two matches. B (red) could be strontium chloride or lithium chloride. The two reds are too close to separate by eye, so the color is a clue, not proof. Accept another test or comparing B side by side with both known samples. Calcium nitrate is also accepted as a second match.
3. A mismatch, with a cause. Accept a real procedural cause tied to what they saw: recording after the wood started burning (yellow-orange), a re-dipped or touched stick bringing in sodium (yellow over lilac), holding the tip too low or too high in the flame (faint color). For cobalt or iron, describing the color they saw and saying it is not on the reference table is enough. "Human error" earns nothing.

GRADING

Suggested split, PROVISIONAL until you set it: pre-lab 8 (1 each) · data table 12 (1 per row with a specific color) · post-lab 6 (2 each). Carbonless copy is graded individually.

MISSED LAB

If the setup is already torn down, the student completes a Missed Lab Analysis rather than the class re-running the lab. Hand them a real class data table from this period (all twelve rows, unknowns labeled A to C only) and have them answer the pre-lab questions 1 to 3 and all three post-lab questions from it.